



Research paper

HFIENet: Hierarchical feature information exchange network for infrared and visible image fusion

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ARTICLE INFO

Keywords:

Image fusion
Infrared and visible image
Information exchange
Hierarchical feature

ABSTRACT

Infrared and visible image fusion (IVIF) is a highly important research field. Through algorithmic fusion, the resulting image contains both the rich texture details present in visible images and the unique thermal information from infrared images. Although numerous approaches have been explored and significant progress has been made, extracting effective modality-specific features and developing robust fusion strategies and rules remain major challenges in IVIF. To address this issue, we propose a novel Hierarchical Feature Information Exchange Network (HFIENet), which comprises two main components: the Information Exchange (IE) module and the Selective Feature Fusion (SFF) module. The IE module employs the cross-attention strategy and differential weighting operation to perform information exchange, thereby enabling the network to extract more significant and comprehensive features from each modality. Due to the different importance of shallow and deep features, the SFF module adaptively integrates essential features for image reconstruction by leveraging attention mechanisms across both channel and spatial dimensions. Extensive experiments conducted on four publicly available datasets demonstrate that HFIENet consistently outperforms current state-of-the-art methods in both qualitative visual analysis and quantitative metric evaluations. Furthermore, under the same experimental settings, it also improves the performance of downstream semantic segmentation and object detection tasks. Our code and pre-trained model will be available at <https://github.com/vonnovx/HFIENet>.

1. Introduction

Given infrared and visible images of the same scene, the objective of IVIF is to generate a single image that conveys more comprehensive scene information than either modality alone. This advantage enables IVIF to maintain strong visual perception even under dynamic or adverse environmental conditions. [1]. As a result, IVIF technology has found wide applicability in a range of advanced computer vision tasks, including target tracking [2], object detection [3], and biometric recognition [4], among others. Over the years, this field has attracted substantial attention from researchers, thereby driving significant advancements in image fusion technology.

Traditional multi-scale transform-based methods [5] and representation learning-based methods [6] can achieve satisfactory fusion results. For instance, Ma et al. [7] employed multi-scale decomposition (MSD) to decompose input images into base layers and detail layers. Subsequently, they performed weighted fusion of the base layers using visual saliency maps (VSM) and optimized the fusion process of the detail layers via weighted least squares (WLS). The generated fusion images exhibit fewer halos, resulting in a more natural visual effect

that aligns better with human perception. Zhou et al. [8] proposed a novel fusion framework that transforms image intensities into the visual response space of the human visual system, enabling feature fusion under a unified perceptual state and significantly improving fusion quality. However, their dependence on handcrafted feature extraction and manually designed fusion rules constrains both the efficiency of the fusion process and the diversity of extracted features.

Deep learning techniques offer significant advantages over traditional approaches owing to their high efficiency and superior generalization capabilities. These methods can be broadly categorized into four main classes: autoencoder (AE)-based methods [9], convolutional neural network (CNN)-based methods [10], generative adversarial network (GAN)-based methods [11], and Transformer-based methods [12]. AE-based methods employ an encoder to extract feature representations from input images, fuse these features through manually defined fusion rules, and then reconstruct the final image using a decoder. CNN-based methods produce fused images in an end-to-end manner under the guidance of well-defined loss functions. GAN-based methods tackle fusion tasks through an adversarial framework consisting of a generator and

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<https://doi.org/10.1016/j.infrared.2026.106428>

Received 29 April 2025; Received in revised form 8 December 2025; Accepted 25 January 2026

Available online 27 January 2026

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a discriminator. Transformer-based methods exploit self-attention or cross-attention mechanisms to model long-range dependencies among image features, thereby strengthening feature extraction and fusion performance. Recently, Li et al. [13] introduced a novel cross-attention mechanism. Unlike most approaches that apply cross-attention to establish inter-modality correlations, their method emphasizes capturing inter-modality irrelevance, which is particularly critical for IVIF tasks. However, the use of cross-attention in their work is confined to the feature fusion stage and is not extended to the feature extraction stage, which limits the exploration of complementary features.

Moreover, existing fusion methods continue to face several challenges. One key challenge lies in figuring out how to optimize feature extraction architectures to capture complementary information from source images more effectively. Another challenge lies in determining how to efficiently process cross-hierarchical features to ensure the completeness of the fused information.

To address these challenges, we propose a CNN-based information exchange network termed HFIENet. Unlike most methods that either adopt a two-stage training strategy (training auto-encoders first, then the fusion module) or extract features via deep network architectures and perform fusion at a single bottleneck, HFIENet is designed as a one-stage paradigm with progressive information exchange. This design enables information exchange across multiple semantic levels during the feature extraction phase. Furthermore, centered on irrelevant complementary features in the feature interaction phase, the method breaks free from the overemphasis on relevant features in previous approaches and leverages a differential weighting mechanism to preserve discriminative information, thereby achieving strong integration with the progressive fusion framework.

Specifically, we develop a step-by-step interactive fusion framework based on residual networks, within which an Information Exchange (IE) module is embedded at the feature extraction stage to facilitate feature interaction. The IE module enhances modality-specific features by introducing the resoftmax activation function [13], subsequently injecting cross-modal information into each other's feature streams through differential weighting. This process not only suppresses extraneous noise, providing a more complete representation for subsequent feature extraction, but also enables preliminary fusion.

In the image reconstruction stage, to fully exploit the fine-grained texture details contained in shallow features and the semantic information embedded in deep features, we introduce a Selective Feature Fusion (SFF) module. This module adaptively integrates the preliminary multi-level features using attention mechanisms across both channel and spatial dimensions. In addition, skip connections are incorporated into high-level features to minimize information loss during reconstruction. In summary, the main contributions of this paper are as follows:

- We propose a CNN-based infrared and visible image fusion network, HFIENet, which extracts and exchanges distinct features across multiple hierarchical levels, enabling the fused results to comprehensively preserve complementary information from both modalities.
- We introduce a novel cross-attention mechanism to enhance the complementarity between modalities. A difference-weighting operation is further employed to inject complementary information into the counterpart feature stream, enabling the network to more comprehensively extract modality-specific features while preserving complementary information.
- To effectively address the varying importance of features across different hierarchical levels, we design a Selective Feature Fusion (SFF) module, which adaptively integrates multi-level fused features through a joint attention mechanism across channel and spatial dimensions.
- Extensive qualitative and quantitative experiments on four publicly available datasets demonstrate that the proposed method achieves superior fusion quality and strong generalization ability while also significantly enhancing the performance of downstream semantic segmentation and object detection tasks.

2. Related work

2.1. AE-based methods

AE-based IVIF methods are generally constructed using an encoder-decoder architecture, where the primary objective is to minimize the loss between the fused image and the input image pair. In the early stages of this research, Li et al. [9] proposed a densely connected network in the encoding stage to improve feature extraction. They employed element-wise addition and the L_1 -norm as fusion strategies to combine the extracted features. Subsequently, recognizing the advantages of multi-scale features for enhancing detail preservation, they introduced a residual network into the fusion pipeline and developed a novel architecture, RFN-Nest [14]. In another study, Liu et al. [15] proposed a deep multi-scale feature integration framework guided by edge attention to better preserve edge information in the fused results. Zhao et al. [16] designed an innovative AE-based structure in which the encoder decomposes the source images into background components (capturing low-frequency information) and detail components (containing high-frequency information), which are then fused independently to retain both global structure and fine textures. Furthermore, Zhao et al. [17] adopted algorithm unrolling techniques to convert iterative optimization algorithms into trainable network architectures. This hybrid approach, which combines model-driven and data-driven paradigms, significantly enhances the efficiency of the fusion process. Recently, Xia et al. [18] proposed DSAFusion for infrared and low-light visible images. It uses a dual-branch encoder to extract detail and semantic features, which are then fused using different networks. This approach achieves a balanced preservation of detail and semantic information.

2.2. CNN-based methods

CNN-based image fusion algorithms typically integrate feature extraction, fusion, and image reconstruction into an end-to-end pipeline, where the design of an effective network architecture and appropriate loss functions plays a pivotal role. For instance, Hou et al. [19] employed an unsupervised CNN-based framework that performs feature extraction and image reconstruction without the need for ground truth supervision. To enhance the representational power of CNN, Long et al. [20] incorporated residual learning and dense connectivity, thereby improving the network's ability to capture and utilize features. Furthermore, Li et al. [21] introduced a low-rank representation-based framework, reformulating feature extraction as an iterative procedure, which not only enhances performance but also improves the interpretability of the model. Tang et al. [22] proposed a light-aware loss function that guides the fusion process by evaluating scene illumination conditions, thus enabling the network to adaptively preserve salient features under varying lighting environments. More recently, Shi et al. [23] integrated Fourier transform analysis into the network, enabling the joint utilization of spatial and frequency domain information from source image pairs and providing a novel and effective strategy for IVIF.

2.3. GAN-based methods

Leveraging the fact that image fusion lacks ground-truth labels, Ma et al. [24] pioneered the use of generative adversarial networks (GAN) in image fusion. Their approach improves fusion quality through an adversarial game between the source images and the fused output. Specifically, the generator is designed to produce fused images with prominent target information, while the discriminator ensures that the fused images retain rich texture details from the visible image. However, relying on a single discriminator can result in modality bias and potential information loss. To address this, Xu et al. [25] proposed DDcGAN, which employs dual discriminators to maintain a

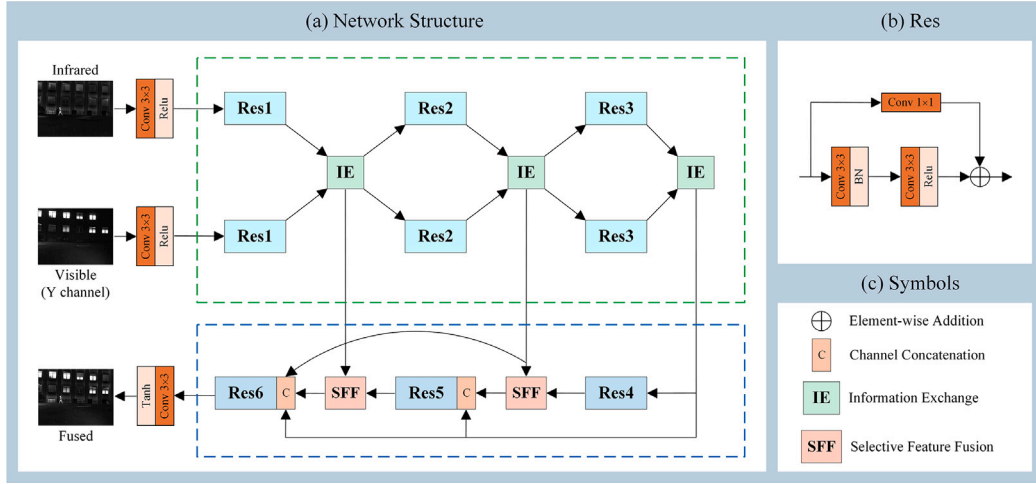


Fig. 1. (a) Our proposed framework for Hierarchical Feature Information Exchange Network. (b) Residual block for different modal feature extraction. (c) Thorough overview of symbols and operations in the network.

balance in the similarity of the fused result to both infrared and visible modalities. Further advancing this idea, Zhou et al. [26] designed a dual-discriminator structure that guides the fusion process using semantic supervision, thereby ensuring semantic consistency between the fused image and the source images. Additionally, Li et al. [27] incorporated a multi-scale attention mechanism into the GAN framework. By introducing an attention-based loss, their method encourages the network to focus on preserving the most informative features during the fusion process.

2.4. Transformer-based methods

Centered on the self-attention mechanism, the Transformer has been widely adopted in image fusion tasks due to its powerful representational capabilities. For instance, Ma et al. [12] integrated the Swin Transformer with a cross-attention mechanism to enable cross-modal learning. Tang et al. [28] introduced YDTR, which combines convolutional neural networks (CNN) and Transformers in both the encoder and decoder branches to leverage their respective strengths. Zhao et al. [29] proposed a dual-branch Transformer-CNN framework that separately extracts the base and detail components of source images. Further, Li et al. [13] proposed a novel cross-attention mechanism that employs an inverse softmax activation function to enhance the complementary features across modalities. Wang et al. [30] formulated IVIF fusion objectives via natural language, which avoids complex, ambiguous mathematical loss modeling and better aligns with human intuitive demands for fusion performance. Recently, Li et al. [31] developed an adversarial attack-resilient network, integrating Transformers to boost feature processing robustness and ensure high-fidelity fused images. Additionally, Li et al. [32] proposed a framework for unaligned multimodal image fusion, which leverages Transformer to enhance global feature extraction and mitigate modality discrepancies between cross-modal images, thereby enabling the seamless integration of registration and fusion tasks within a single-stage framework.

3. Method

3.1. Framework overview

HFINET is a CNN-based, end-to-end network architecture, as illustrated in Fig. 1. It comprises three primary components: residual block (Res), information exchange (IE), and selective feature fusion (SFF).

Let the infrared and visible images be denoted by $I_{ir} \in \mathbb{R}^{1 \times H \times W}$ and $I_{vis} \in \mathbb{R}^{3 \times H \times W}$, respectively, where H and W represent the height and

width of the image, and C denotes the number of feature channels. The visible image is first converted from the RGB color space to the YCbCr color space, and then the Y channel is extracted as the input. The preprocessed visible and infrared images are passed through a 3×3 convolutional layer followed by ReLU activation to generate the initial feature maps F_{ir}^0 and F_{vis}^0 . Subsequently, modality hierarchical features are extracted by Res blocks arranged along two independent branches. These independent branches ensure that features from each modality are rich and distinctive. The structure of the Res block is shown in Fig. 1(b). It adopts a residual design to alleviate vanishing or exploding gradients during training and to enhance the network's feature extraction capacity. Notably, the feature extraction modules use non-shared weights to maintain cross-modal specificity. This process is described by:

$$\begin{aligned} F_{ir}^i &= \mathcal{R}^i(\hat{F}_{ir}^{i-1}) \\ F_{vis}^i &= \mathcal{R}^i(\hat{F}_{vis}^{i-1}), \quad i = 1, 2, 3 \end{aligned} \quad (1)$$

where $\mathcal{R}^i(\cdot)$ represents the i th Res block, F_{ir}^i and F_{vis}^i are the extracted hierarchical features, and $\hat{F}_{ir}^{i-1}$, $\hat{F}_{vis}^{i-1}$ are the initial feature maps ($i = 1$) or the modal features after information exchange ($i = 2, 3$).

As our network incrementally extracts features across multiple stages, each Res module is followed by an Information Exchange (IE) module. This module facilitates cross-modal complementary feature interaction, enhancing feature representation for subsequent extraction stages while also serving as an initial fusion mechanism. The process is defined as:

$$(\hat{F}_{ir}^i, \hat{F}_{vis}^i) = \mathcal{I}^i(F_{ir}^i, F_{vis}^i) \quad (2)$$

$$F_f^i = \text{Conv}(\text{Concat}(\hat{F}_{ir}^i, \hat{F}_{vis}^i)) \quad (3)$$

where $\mathcal{I}^i(\cdot)$ denotes the i th IE module, Concat refers to channel-wise concatenation, and Conv is a 3×3 convolutional layer used to compress the channel dimension of the fused features. F_f^i represents the pre-fused features of the i th layer.

Following feature extraction and initial fusion, the hierarchical fused features are forwarded to the reconstruction network, where they undergo adaptive fusion in a progressive manner. This process enables the network to learn the relative importance of features at each level and to utilize them effectively for final image reconstruction. The operations within the Selective Feature Fusion (SFF) module are formulated as:

$$\hat{F}_f^i = \mathcal{S}^i(F_f^i, F_{out}^{i+1}) \quad (4)$$

Table 1

Number of input channels and output channels for different residual blocks.

Module	Input channels	Output channels
Res1	8	16
Res2	16	32
Res3	32	64
Res4	64	32
Res5	96	16
Res6	112	8

where $S^i(\cdot)$ denotes the i th SFF module and F_{out}^{i+1} represents the fusion feature processed by the $i+1$ -th residual block during the reconstruction process.

To mitigate information loss during reconstruction, skip connections are introduced between deep layers. After processing through multiple Res blocks, the final feature map is obtained and passed through a 3×3 convolutional layer followed by a hyperbolic tangent activation function to generate the final fused image:

$$I_f = \text{Tanh}(\text{Conv}(F_f^0)) \quad (5)$$

where F_f^0 denotes the final feature map and I_f is the reconstructed fused image. We further summarize the entire information processing flow of HFIENet and more architectural details of the residual modules in Algorithm 1 and Table 1.

Algorithm 1 Hierarchical Feature Information Exchange Network

Input: Infrared image I_{ir} , Visible image I_{vis}

Output: Fused image I_f

- 1: **Preprocessing:**
 - 2: Convert I_{vis} to YCbCr color space, and extract the Y channel: I_{vis}^y
 - 3: $F_{ir}^0 \leftarrow \text{ReLU}(\text{Conv}_{3 \times 3}(I_{ir}))$
 - 4: $F_{vis}^0 \leftarrow \text{ReLU}(\text{Conv}_{3 \times 3}(I_{vis}^y))$
 - 5: Let $\hat{F}_{ir}^0 = F_{ir}^0$, $\hat{F}_{vis}^0 = F_{vis}^0$
 - 6: **Hierarchical Feature Extraction and Exchange:**
 - 7: **for** $i = 1$ to 3 **do**
 - 8: $F_{ir}^i \leftarrow \mathcal{R}^i(\hat{F}_{ir}^{i-1})$
 - 9: $F_{vis}^i \leftarrow \mathcal{R}^i(\hat{F}_{vis}^{i-1})$
 - 10: $(\hat{F}_{ir}^i, \hat{F}_{vis}^i) \leftarrow \mathcal{I}^i(F_{ir}^i, F_{vis}^i)$
 - 11: $F_f^i \leftarrow \text{Conv}_{3 \times 3}(\text{Concat}(\hat{F}_{ir}^i, \hat{F}_{vis}^i))$
 - 12: **end for**
 - 13: **Progressive Fusion and Reconstruction:**
 - 14: $F_{out}^3 \leftarrow \mathcal{R}^3(F_f^3)$
 - 15: $\hat{F}_f^2 \leftarrow S^2(F_f^2, F_{out}^3)$
 - 16: $F_{out}^2 \leftarrow \mathcal{R}^2(\text{Concat}(\hat{F}_f^2, F_f^3))$
 - 17: $\hat{F}_f^1 \leftarrow S^1(F_f^1, F_{out}^2)$
 - 18: $F_{out}^1 \leftarrow \mathcal{R}^1(\text{Concat}(\hat{F}_f^1, F_f^2, F_f^3))$
 - 19: **Final Image Generation:**
 - 20: Let $F_f^0 = F_{out}^1$
 - 21: $I_f \leftarrow \text{Tanh}(\text{Conv}_{3 \times 3}(F_f^0))$
-

3.2. IE architecture

As previously discussed, the objective of IVIF is to produce a fused image that retains as much complementary information as possible from both the infrared and visible source images. Nevertheless, how to effectively extract and utilize such complementary information within end-to-end fusion networks remains insufficiently explored. To address this issue, we design an Information Exchange (IE) module, as illustrated in Fig. 2, which consists of two stages: complementary attribute enhancement and differential information exchange. First, we adopt a cross-attention mechanism with an unconventional softmax

function to enhance the inter-modal complementary attributes from a global perspective, thereby preventing complementary information from being overshadowed by correlated features. Subsequently, we focus on the most salient features in the differential feature maps through element-wise subtraction and global max-pooling operations to reduce noise interference. These refined differential features are injected into each other's feature streams via channel weighting. This process not only achieves preliminary feature fusion but also provides a more comprehensive information representation for subsequent feature extraction.

In the first stage, we perform global complementary attribute enhancement between the two branches by introducing a novel resoftmax [13] activation function within our cross-attention mechanism. Specifically, we adapt the Multi-Dconv Head Transposed Attention (MDTA) block from Restormer [33], where the Query is derived from the other modality and softmax is replaced with resoftmax. This modification enables cross-modal attention that effectively captures complementary features while reducing computational complexity. The computation is as follows:

$$[Q_{\hat{c}}, K_c, V_c] = W_{qkv}^d W_{qkv}^p [F_{\hat{c}}, F_c, F_c] \quad (6)$$

$$U = V_c \text{resoftmax} \left(\frac{K_c^T Q_{\hat{c}}}{\alpha} \right) \quad (7)$$

where, $F_{\hat{c}}$ and F_c represent the features from different modalities; W_{qkv}^p and W_{qkv}^d denote 1×1 point-wise and 3×3 depth-wise convolution kernels, respectively. α is a learnable scaling parameter, and U is the resulting cross-attention output. The resoftmax function is defined as:

$$\text{resoftmax}(X) = \text{softmax}(-X) \quad (8)$$

As shown in Fig. 2, resoftmax produces an inverse trend compared to softmax, where larger input values correspond to smaller outputs. This encourages the network to focus on regions of lower similarity, thus enhancing attention to modality-specific texture details and promoting complementary feature extraction.

In the second stage, inspired by the principle of differential amplification [34], we generate differential feature maps through element-wise subtraction between the complementary-enhanced modality features:

$$\begin{aligned} F_{ir}^A &= F_{ir}' - F_{vis}' \\ F_{vis}^A &= F_{vis}' - F_{ir}' \end{aligned} \quad (9)$$

where F_{ir}' and F_{vis}' denote infrared and visible features enhanced by cross-attention.

To further refine these differential features, global max pooling (GMP) is applied to emphasize high-frequency details and suppress irrelevant noise. The pooled features are then passed through a series of convolutional layers and activation functions for dimensional adjustment, followed by sigmoid normalization. The resulting channel-wise weights are employed to modulate the complementary features, which are subsequently injected into the counterpart feature stream via element-wise addition:

$$\begin{aligned} \hat{F}_{ir} &= F_{ir} + F_{vis}^A \cdot \sigma(\text{Convs}(\text{GMP}(F_{vis}^A))) \\ \hat{F}_{vis} &= F_{vis} + F_{ir}^A \cdot \sigma(\text{Convs}(\text{GMP}(F_{ir}^A))) \end{aligned} \quad (10)$$

where GMP denotes global max pooling, Convs represents the convolutional layers and non-linear activations, and σ denotes the sigmoid function.

Finally, to obtain the pre-fusion features at each level, we concatenate the exchanged features along the channel dimension and apply a 3×3 convolutional layer to unify the output channel size. These preliminary fusion features are used for subsequent image reconstruction.

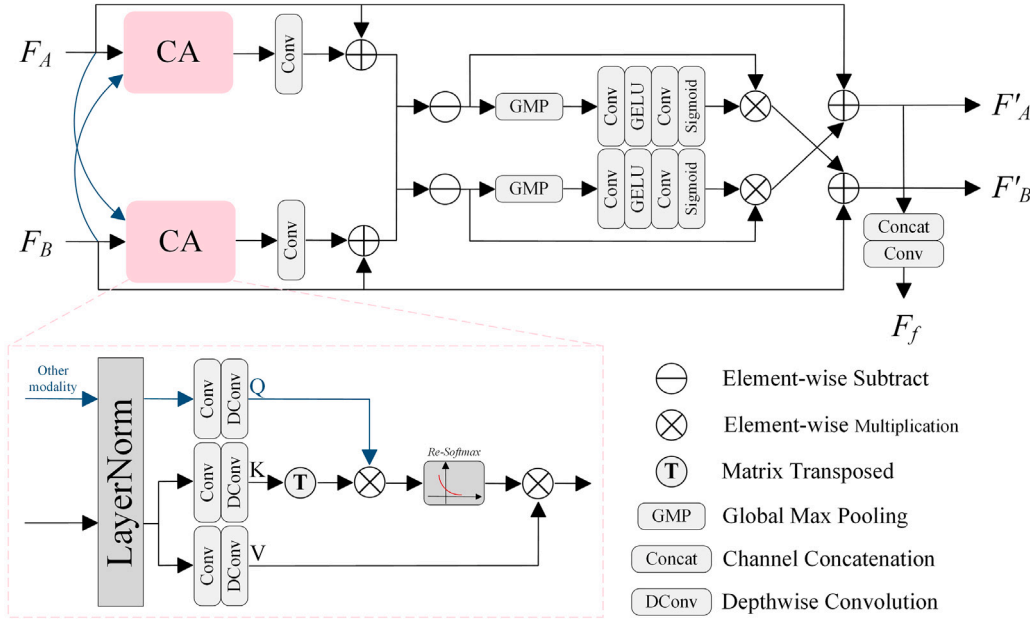


Fig. 2. Architecture of the IE.

3.3. SFF architecture

The generation of a high-quality fused image demands the preservation of fine texture details from shallow features alongside high-level semantic information from deeper representations. However, increasing the network depth during feature extraction often leads to the degradation or loss of shallow features. This presents a significant challenge in effectively utilizing fusion features at various depths. While simple concatenation or summation is commonly used, these operations are limited in their ability to fully exploit the representational capacity of the network.

To overcome this limitation, we design a Selective Feature Fusion (SFF) module, which replaces fixed rule-based fusion with adaptive dynamic selection of hierarchical features to address the varying demands of different feature levels. By combining channel and spatial attention mechanisms, the module adaptively integrates features in a progressive manner, prioritizes the retention of important information, and minimizes loss during image reconstruction. As illustrated in Fig. 3, given the input shallow and deep features, feature aggregation is first performed using channel-wise concatenation and element-wise addition to generate F_c and F_a , respectively. Channel self-attention (CSA) is then applied to F_c to capture global dependencies among feature channels.

Local spatial dependencies are first modeled by applying a 1×1 convolution to F_a , followed by a sigmoid activation to generate spatial attention weights. These weights are then used to calibrate the global channel features by emphasizing critical spatial regions, resulting in a multi-scale feature descriptor. A subsequent sigmoid activation further produces a soft selection, allowing the network to adaptively integrate features across different hierarchical levels. The overall process is formulated as follows:

$$F_c = \text{Concat}(F_{f_s}, F_{f_d}) \quad (11)$$

$$F_a = F_{f_s} + F_{f_d} \quad (12)$$

$$\omega = \sigma(\text{CSA}(F_c)) \cdot \sigma(\text{Conv}(F_a)) \quad (13)$$

$$\hat{F}_f = F_{f_d} \cdot \omega + F_{f_s} \cdot (1 - \omega)$$

where F_{f_s} and F_{f_d} represent the shallow and deep features, respectively; Conv denotes a 1×1 convolutional layer; and CSA stands for the channel self-attention operation.

3.4. Loss function

To ensure that the fusion results generated by HFIENet contain both rich detail and salient information, we define the total loss function as follows:

$$\mathcal{L}_{total} = \alpha \cdot \mathcal{L}_{grad} + \beta \cdot \mathcal{L}_{pixel} \quad (14)$$

where $\mathcal{L}_{grad}$ and $\mathcal{L}_{pixel}$ denote the gradient loss and pixel loss, respectively. The coefficients α and β are hyperparameters that control the relative contributions of the two losses.

The gradient loss encourages the preservation of important texture details in the fused image. It is computed as:

$$\mathcal{L}_{grad} = \frac{1}{HW} \|\nabla I_f - \max(|\nabla I_{ir}|, |\nabla I_{vis}|)\|_1 \quad (15)$$

where H and W are the height and width of an image, respectively, ∇ denotes the Sobel operator for gradient computation, and $\|\cdot\|_1$ represents the l_1 -norm.

In order to retain both essential content and salient regions in the fused result, we adopt a segmented pixel loss strategy as proposed in [35]. This method first evaluates the importance of each pixel using the product of its intensity and its gradient magnitude:

$$p_i = \nabla I_s \cdot I_s \quad (16)$$

where, I_s represents the source image.

Pixels are then divided into two subsets based on their importance scores:

$$\begin{aligned} p_{part1} &= \{p_i | (p_i \geq p_{ir}^\tau \text{ or } p_i \geq p_{vis}^\tau)\} \\ p_{part2} &= \{p_i | (p_i < p_{ir}^\tau \text{ and } p_i < p_{vis}^\tau)\} \end{aligned} \quad (17)$$

where p_{vis}^τ and p_{ir}^τ represent the top $\tau\%$ of the most important pixels in the visible and infrared images, respectively.

For the highly informative pixels in p_{part1} , the loss is defined with respect to the maximum value between the two source modalities:

$$\mathcal{L}_{part1} = \frac{1}{HW} \|I_f - \max(I_{ir}, I_{vis})\|_1, \quad p_i \in p_{part1} \quad (18)$$

For the remaining less significant pixels (p_{part2}), an average of the two modalities is used:

$$\mathcal{L}_{part2} = \frac{1}{2HW} (\|I_f - I_{ir}\|_1 + \|I_f - I_{vis}\|_1), \quad p_i \in p_{part2} \quad (19)$$

The final pixel loss is defined as the sum of the two components:

$$\mathcal{L}_{pixel} = \mathcal{L}_{part1} + \mathcal{L}_{part2} \quad (20)$$

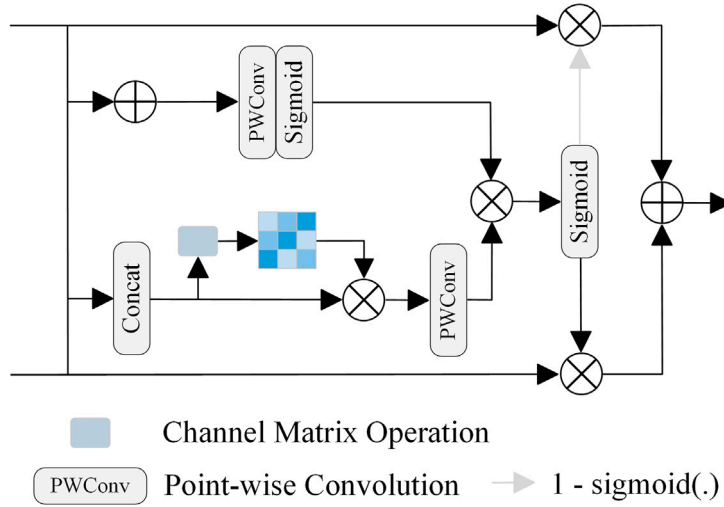


Fig. 3. Architecture of the SFF.

4. Experiments

4.1. Experimental configurations

To evaluate the performance of HFNet, we conduct extensive comparative experiments against nine state-of-the-art methods across four widely used datasets: MSRS [22], TNO [36], RoadScene [37], and M3FD [11]. The compared methods include U2Fusion [37], MFEIF [15], YDTR [28], IRFS [38], DCINN [39], CrossFuse [13], DATFuse [40], TUFusion [41], and LRRNet [21]. To ensure fairness in the comparison, all comparison methods are implemented using their publicly available source code, and we adopt their default configurations without any modifications.

To conduct a comprehensive and multidimensional quantitative evaluation of the fusion results, we select eight widely recognized metrics in the field of image fusion, namely Entropy (EN), Standard Deviation (SD), Spatial Frequency (SF), Mutual Information (MI), Sum of the Correlations of Differences (SCD), Visual Information Fidelity (VIF), Qabf, and Average Gradient (AG), with their calculation details available in [42]. The rationale for choosing these metrics lies in the fact that no single metric can fully assess the quality of a fused image. Instead, this set of metrics enables a more objective and holistic evaluation of fusion performance from multiple perspectives, including information content, statistical characteristics, texture and detail preservation, information fidelity, and human visual perception.

EN quantifies the amount of information contained in the fused image from the perspective of information theory. SD evaluates image contrast and intensity distribution based on statistical characteristics. SF reveals the details and texture of the fused image by computing the gradient distribution. MI, also derived from information theory, measures how much information the fused image inherits from the source images. SCD assesses fusion quality by calculating the correlation between the difference images of the fused image and each source image. VIF estimates image fidelity from the perspective of human visual perception. Qabf measures the preservation of edge information. AG reflects the richness of textures by analyzing gradient variations. Higher values across all these metrics indicate superior fusion performance.

4.2. Implementation details

We train our model using the training set (1083 pairs) from the MSRS dataset and evaluate it on the corresponding test set (361 pairs). For data augmentation, all training images are randomly cropped into 128×128 patches. To further assess the generalization capability of our

approach beyond the MSRS dataset, we conduct additional evaluations on the TNO (25 pairs), RoadScene (50 pairs), and M3FD (50 pairs) datasets.

The proposed method is implemented on a machine equipped with an NVIDIA GeForce RTX 4090 GPU using the PyTorch framework. During training, we employ the Adam optimizer with an initial learning rate of 1.6×10^{-4} , which is reduced by a factor of 0.5 every 10 epochs. The batch size is set to 8, and the model is trained for a total of 50 epochs. The hyperparameters for the loss function are configured as $\alpha = 20$ and $\beta = 20$.

4.3. Fusion experiment analysis

4.3.1. Qualitative comparison

The visual comparisons of various fusion methods across the four datasets are illustrated in Fig. 4, 5, 6, and 7. To facilitate a more intuitive assessment of their differences, we enlarge specific target regions and detailed areas.

For the MSRS dataset, representative daytime and nighttime scenes are selected for evaluation. As shown in Fig. 4, most methods demonstrate satisfactory fusion performance, successfully integrating complementary information from the source images to varying degrees. However, subtle differences remain between the methods. In daytime scenes, approaches such as U2Fusion, YDTR, and CrossFuse exhibit subpar performance in preserving character saliency and object detail. These methods fail to extract sufficient salient information from the infrared (IR) images, resulting in low contrast and diminished clarity. Meanwhile, TUFusion and LRRNet tend to suppress edge information, leading to blurred contours and residual artifacts. By contrast, DCINN, DATFuse, and our proposed method better preserve texture detail, with our approach yielding superior visual quality. In nighttime scenarios, where salient edge information is critical for character target recognition, our method excels in balancing overall brightness and saliency. Unlike other methods that struggle to harmonize these aspects, our fused results effectively highlight salient features while maintaining natural luminance, thereby demonstrating robust integration of complementary information.

Across diverse scenes depicted in Fig. 5, IR images distinctly render human figures due to thermal radiation. However, methods such as U2Fusion, CrossFuse, and LRRNet diminish the brightness of high-luminance targets in the IR domain. Notably, in the second scene, where a character is partially occluded by tree branches, only our method is able to clearly render the figure's features. Regarding the green-highlighted regions containing tree branches, methods like MFEIF, DCINN, DATFuse, and TUFusion exhibit varying degrees of

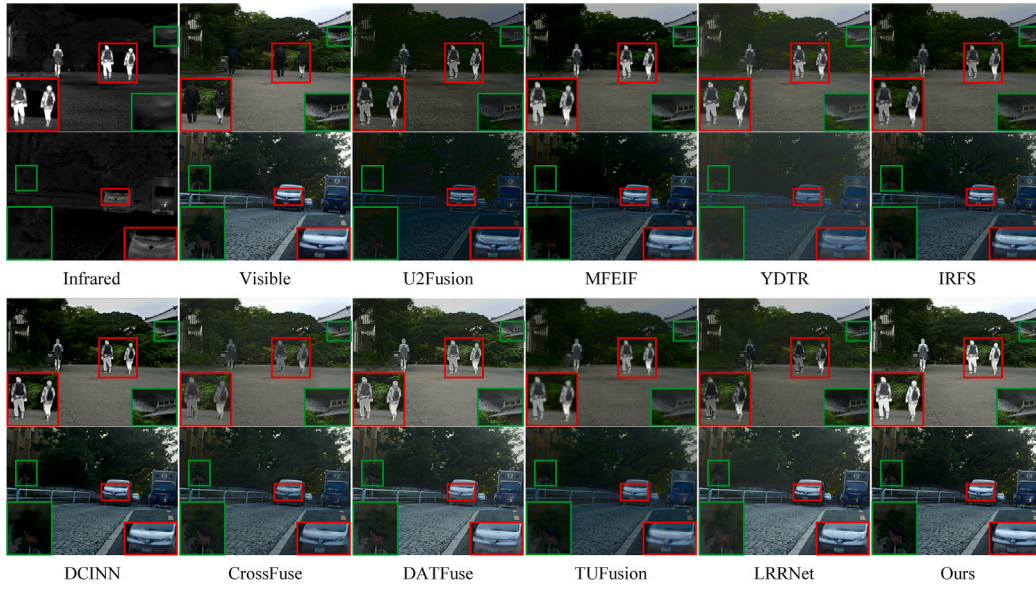


Fig. 4. Qualitative comparison results of HFIENet against nine other methods on the MSRS dataset.

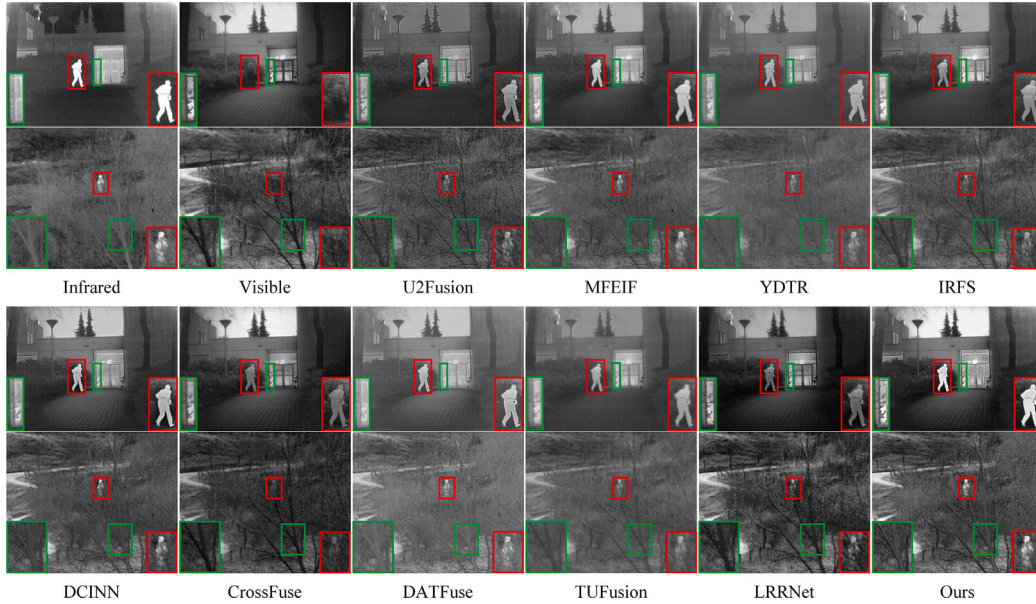


Fig. 5. Qualitative comparison results of HFIENet against nine other methods on the TNO dataset.

boundary blurring and insufficient detail preservation. In contrast, our method maintains well-defined contours and textures of tree trunks and foliage in both foreground and background areas, while simultaneously achieving a balanced representation of target saliency and fine-grained details.

In the first scene of Fig. 6, the bright sky background presents a challenge for detail preservation. Only CrossFuse and our method successfully render the wire structures, whereas U2Fusion, IRFS, and TUFusion fail to distinguish the sky from the wires, resulting in low contrast. Character and vehicle features in the background, which are prominent in the IR image, appear in most fusion outputs. However, our approach provides a more faithful representation of character outlines and vehicle textures. In the second scene, methods such as MFEIF, YDTR, DCINN, and DATFuse produce blurred representations of the figures, and the barbed wire textures are inadequately preserved. From the perspective of scene coherence and visual fidelity, our method consistently achieves superior results.

Moreover, as clearly illustrated in Fig. 7, our network effectively preserves the distinctive characteristics of human figures in the first scene and the intricate structural details of the vehicle wheels in the second scene, resulting in fused images with high contrast and enriched visual information. This superior performance stems from the proposed architecture's capacity to exploit multi-level features and dynamically integrate complementary information from both infrared and visible modalities.

4.3.2. Quantitative comparison

The quantitative evaluation results across four benchmark datasets are summarized in the table. As shown in Table 2, our method consistently achieves superior performance across multiple metrics on the MSRS test set, outperforming all compared approaches. The SD scores are significantly higher, suggesting that the fused images produced by our method exhibit superior contrast. This quantitative result is consistent with the findings of the preceding qualitative analysis. Elevated

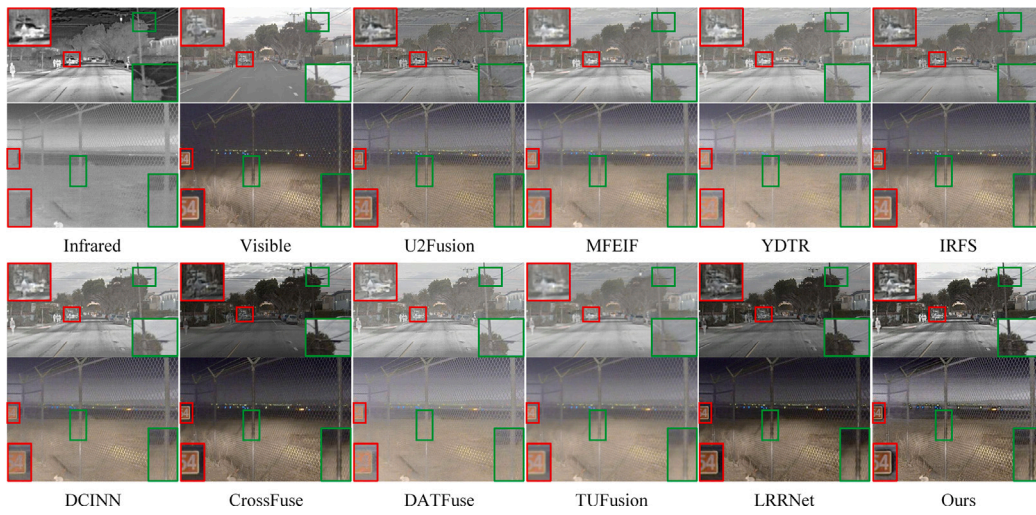


Fig. 6. Qualitative comparison results of HFIENet against nine other methods on the RoadScene dataset.

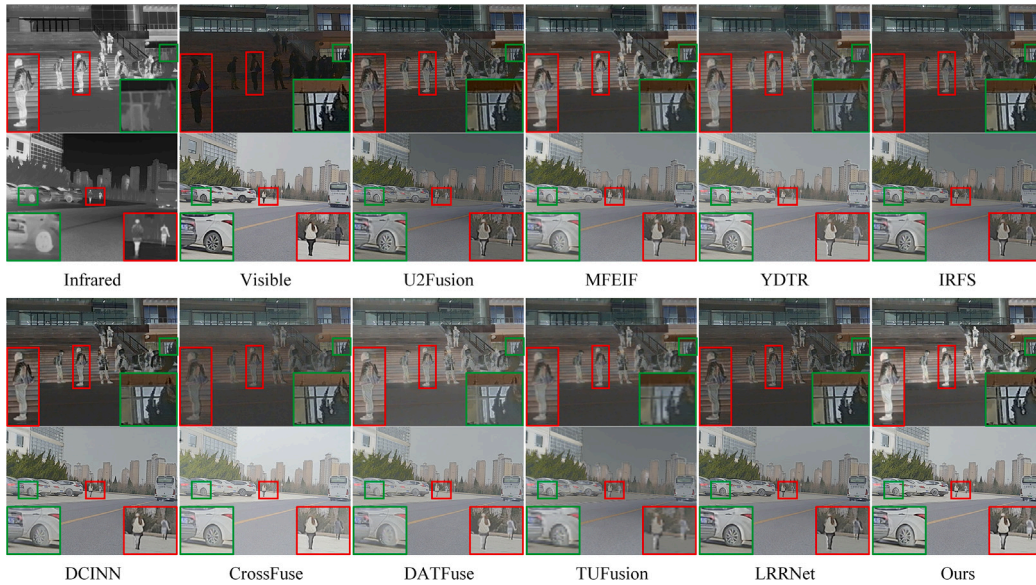


Fig. 7. Qualitative comparison results of HFIENet against nine other methods on the M3FD dataset.

Table 2

The quantitative comparison results between HFIENet and nine other methods on the MSRS dataset. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	EN	SD	SF	MI	SCD	VIF	Qabf	AG
U2Fusion	5.04	20.34	7.24	1.37	1.07	0.49	0.34	2.26
MFEIF	5.77	33.82	8.08	2.06	1.52	0.75	0.55	2.60
YDTR	5.65	25.37	7.40	1.92	1.14	0.58	0.35	2.19
IRFS	6.28	31.18	8.61	1.95	1.49	0.77	0.54	2.71
DCINN	6.00	40.29	10.51	2.30	1.49	0.82	0.57	3.34
CrossFuse	6.50	36.44	9.66	2.17	1.05	0.84	0.56	3.01
DATFuse	6.48	36.48	10.93	2.70	1.41	0.91	0.64	3.56
TUFusion	6.08	26.40	4.64	1.78	1.30	0.57	0.23	1.68
LRRNet	6.19	31.76	8.47	2.03	0.79	0.54	0.45	2.64
Ours	6.69	43.19	11.28	3.33	1.66	1.06	0.71	3.70

EN and MI scores further suggest that our fusion strategy effectively conveys more informative content from the source images. In addition, the high SF and AG values reflect the preservation of abundant texture details in the fused outputs. The top-ranked Qabf score demonstrates our method's ability to retain critical edge information, while the leading SCD result confirms the richness of fused image content. Moreover, the excellent performance in VIF highlights the alignment of our fusion results with human visual perception, ensuring strong visual fidelity.

Furthermore, our method also shows robust generalization capabilities across the TNO, RoadScene, and M3FD datasets. As shown in Table 3, 4, and 5, HFIENet achieves the highest scores in seven out of eight metrics, including EN, SD, SF, MI, VIF, Qabf, and AG, and only slightly trails behind in SCD. Notably, when comparing across different cross-domain datasets, our method consistently outperforms the second-ranked method significantly in terms of SD. A higher SD value indicates a wider grayscale range and higher contrast. In HFIENet,

Table 3

The quantitative comparison results between HFIENet and nine other methods on the TNO dataset. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	EN	SD	SF	MI	SCD	VIF	Qabf	AG
U2Fusion	6.42	26.34	8.85	1.35	1.51	0.54	0.42	3.60
MFEIF	6.71	34.50	7.76	1.83	1.72	0.67	0.47	3.09
YDTR	6.42	28.25	8.01	1.69	1.50	0.60	0.40	2.85
IRFS	6.67	32.13	9.52	1.60	1.70	0.62	0.46	3.50
DCINN	6.80	36.49	10.20	1.98	1.63	0.70	0.49	3.90
CrossFuse	6.89	40.29	10.20	2.18	1.26	0.73	0.46	3.76
DATFuse	6.58	29.65	10.09	2.36	1.45	0.70	0.51	3.74
TUFusion	6.52	28.31	4.68	1.47	1.59	0.46	0.23	2.02
LRRNet	6.90	38.57	9.44	1.76	1.53	0.54	0.37	3.63
Ours	7.08	44.82	12.18	2.82	1.56	0.86	0.56	4.70

Table 4

The quantitative comparison results between HFIENet and nine other methods on the RoadScene dataset. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	EN	SD	SF	MI	SCD	VIF	Qabf	AG
U2Fusion	6.72	29.64	10.29	1.84	1.38	0.56	0.50	4.10
MFEIF	6.96	37.26	8.85	2.29	1.68	0.66	0.50	3.40
YDTR	6.78	33.32	9.68	2.13	1.42	0.61	0.47	3.52
IRFS	6.79	32.40	9.64	1.90	1.63	0.58	0.45	3.52
DCINN	7.00	38.28	11.31	2.21	1.55	0.67	0.52	4.23
CrossFuse	7.12	45.36	10.73	2.34	1.31	0.61	0.37	3.87
DATFuse	6.72	32.27	11.32	2.62	1.29	0.63	0.52	4.00
TUFusion	6.73	30.79	5.87	1.92	1.52	0.52	0.29	2.41
LRRNet	6.99	39.89	11.12	1.99	1.61	0.49	0.35	4.01
Ours	7.28	51.67	15.38	2.84	1.36	0.76	0.51	5.55

Table 5

The quantitative comparison results between HFIENet and nine other methods on the M3FD dataset. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	EN	SD	SF	MI	SCD	VIF	Qabf	AG
U2Fusion	6.69	29.36	13.51	1.61	1.68	0.54	0.56	5.05
MFEIF	6.79	31.21	10.92	2.03	1.73	0.59	0.48	3.91
YDTR	6.75	29.95	13.41	2.15	1.65	0.59	0.52	4.44
IRFS	6.67	29.43	13.42	1.72	1.75	0.59	0.59	4.59
DCINN	6.88	32.46	13.99	2.39	1.58	0.62	0.52	4.82
CrossFuse	6.73	33.36	13.72	2.43	1.05	0.69	0.60	4.59
DATFuse	6.63	27.59	12.58	2.79	1.41	0.56	0.47	4.10
TUFusion	6.47	26.84	5.22	1.63	1.59	0.30	0.12	2.06
LRRNet	6.61	28.49	13.12	1.93	1.60	0.52	0.53	4.46
Ours	7.08	38.59	17.27	3.38	1.48	0.75	0.64	5.80

the IE module simultaneously preserves the dark-region textures of visible images and the bright-region targets of infrared images through cross-attention and differential information exchange, thereby directly expanding the pixel dynamic range. Meanwhile, the SFF module balances shallow and deep features via attention mechanisms, preventing high-frequency details from being smoothed during reconstruction. The synergy between these two modules endows the fused images with a wide grayscale range and high contrast, ultimately leading to the model's excellent performance in the SD metric. These results indicate that our method not only effectively extracts and integrates complementary information from multimodal data but also consistently maintains strong contrast and perceptual quality across diverse scenes, highlighting its effectiveness and competitive advantage in producing fusion results enriched with both texture and semantic content.

4.4. Ablation studies

4.4.1. Ablation of architecture

To assess the effectiveness of the network core module, we perform a series of ablation experiments on the MSRS test set. As illustrated in Fig. 8, HFIENet is decomposed into four architectural variants for comparative analysis. The baseline configuration (Fig. 8(a)) consists of the original framework employing a three-level Res block as the backbone of the feature extraction branch. The configuration in Fig. 8(b) (backbone+IE) introduces the information exchange (IE) module to assess its contribution. Fig. 8(c) (backbone+IE+SFF) further incorporates the selective feature fusion (SFF) module to examine its impact.

Finally, Fig. 8(d) (backbone+IE+SFF+Skip Connections) represents the full HFIENet model, where skip connections are added to evaluate their role in the image reconstruction process. The quantitative results of the ablation study are presented in Table 6. It is evident that the full model (Fig. 8(d)) consistently achieves the highest scores across most metrics. Moreover, as the architectural components are incrementally introduced from Fig. 8(a) to Fig. 8(d), the performance improves progressively, validating the individual and collective effectiveness of each module. Specifically, the IE module significantly enhances the extraction of cross-modal complementary features, while the SFF module enables more efficient integration of hierarchical features. The addition of skip connections further mitigates information loss during reconstruction. The combined use of these components enables the network to generate high-quality fused images.

4.4.2. Effects of different subcomponents

To systematically verify the effectiveness of each core sub-component in the proposed HFIENet, we conduct a series of detailed ablation experiments. These experiments aim to independently evaluate the contributions of the two key stages of the IE module, the attention mechanisms in the SFF module, and the pixel loss function to the final fusion performance. The comparative experimental setups include the following model variants: w/o CA refers to removing the complementary attribute enhancement stage, where original features directly enter the differential information exchange stage; w/o DE means removing the differential information exchange stage, where features enhanced by CA are directly used for subsequent processing

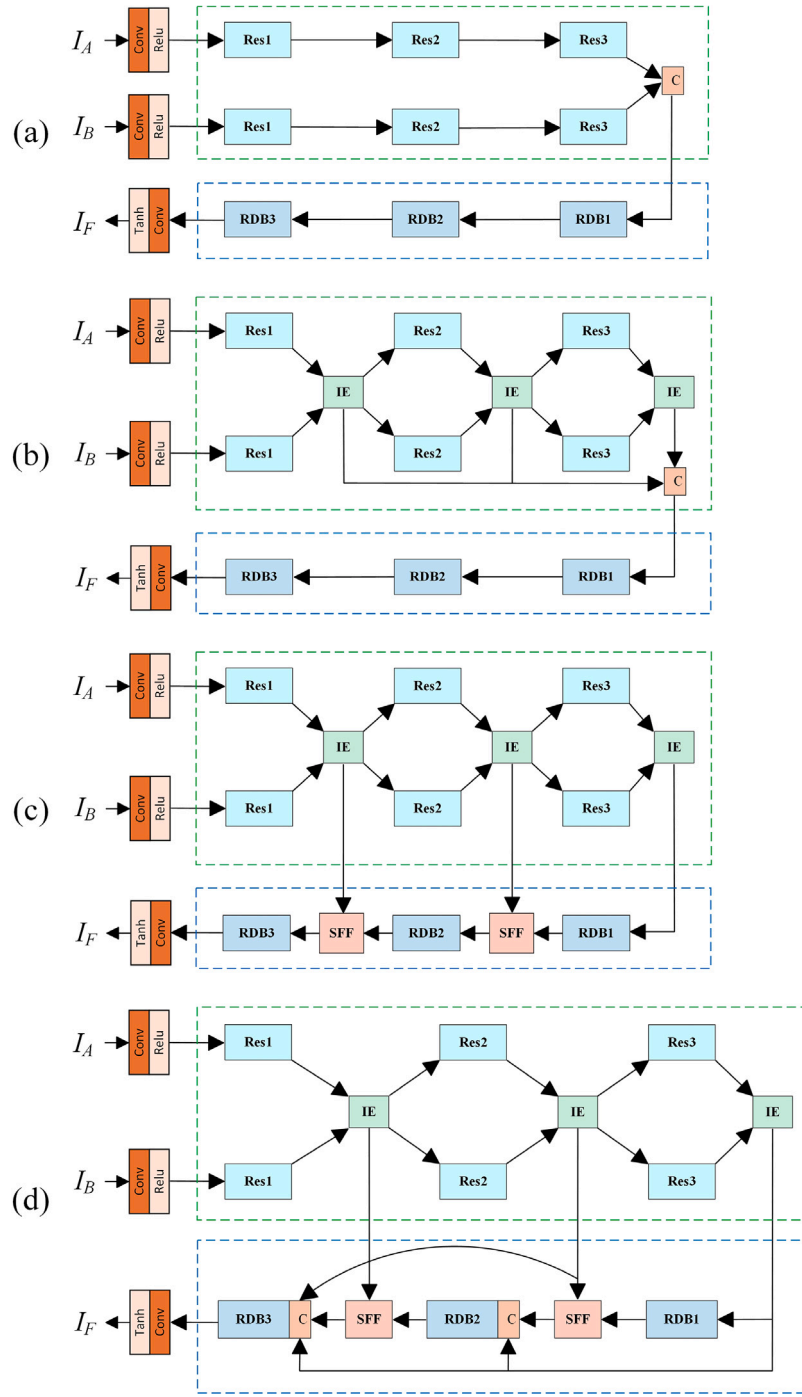


Fig. 8. Four network architecture variants for ablation experiments. (a) Backbone. (b) Backbone + IE. (c) Backbone + IE + SFF. (d) Backbone + IE + SFF + Skip Connections (ours).

Table 6

Performance metrics of network structures combined by different methods. SC indicates Skip Connections. Optimal values are in **red**.

IE	SFF	SC	EN	SD	SF	MI	SCD	VIF	Qabf	AG
✓			6.51	38.13	10.07	3.04	1.51	0.97	0.69	3.32
✓			6.67	42.89	11.23	3.46	1.65	1.05	0.70	3.64
✓	✓		6.68	42.66	11.26	3.45	1.64	1.06	0.71	3.67
✓	✓	✓	6.69	43.19	11.28	3.33	1.66	1.06	0.71	3.70

without differential injection; w/o CSA denotes removing the channel self-attention mechanism from the SFF module; w/o SA represents removing the spatial attention mechanism from the SFF module; w/o

SFF refers to replacing the SFF module with DARM from DATFuse [40]; w/o pixel indicates removing the piecewise-pixel loss function during training. The quantitative evaluation results are presented in Table

Table 7

Quantitative results of ablation experiments assessing the effect of different subcomponents. Optimal values are in red.

	EN	SD	SF	MI	SCD	VIF	Qabf	AG
w/o CA	6.66	42.57	11.26	3.37	1.63	1.05	0.71	3.68
w/o DE	6.64	42.68	11.19	3.38	1.60	1.03	0.70	3.64
w/o CSA	6.66	42.95	11.15	3.37	1.64	1.05	0.70	3.70
w/o SA	6.67	43.12	11.20	3.36	1.65	1.04	0.69	3.66
w/o SFF	6.61	41.54	11.26	2.82	1.51	0.88	0.66	3.68
w/o $\mathcal{L}_{\text{pixel}}$	6.66	42.72	11.25	3.34	1.44	0.92	0.70	3.68
Ours	6.69	43.19	11.28	3.33	1.66	1.06	0.71	3.70

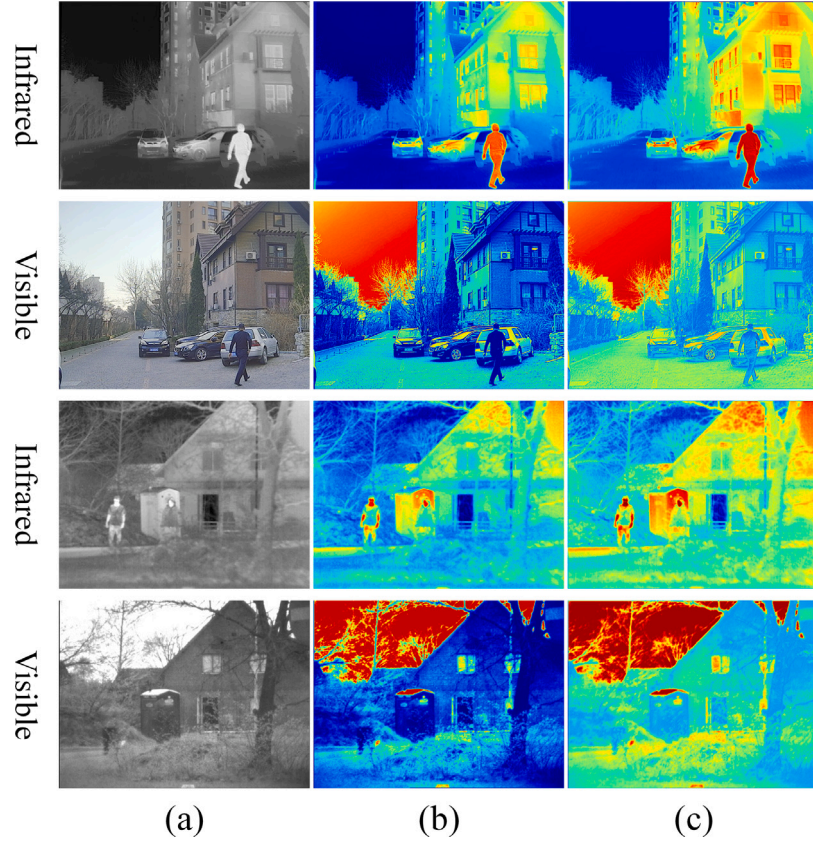


Fig. 9. Feature visualization before and after resoftmax activation. (a) Infrared or visible images. (b) Feature representation before activation. (c) Feature representation after activation.

7. Additionally, we visualize the features following activation by the resoftmax function to intuitively verify its effectiveness in enhancing complementary properties, with the results shown in Fig. 9.

In Table 7, by comparing the w/o CA and w/o DE variants, we can quantify the contributions of the two stages of the IE module. Removing either the CA or DE stage leads to a decline in most metrics of the model. In particular, the removal of the DE stage results in a more significant drop in model metrics, which strongly confirms that our differential information injection mechanism is the core driver for improving fusion performance. Meanwhile, the performance degradation after removing the CA stage also verifies the necessity of pre-enhancing complementary attributes. For the w/o CSA and w/o SA variants, the absence of the channel self-attention mechanism and the spatial attention mechanism, respectively, causes the model to underperform the complete model in metrics such as EN, SD, and VIF. This indicates that the dual-attention design, where CSA captures global channel dependencies between features and SA focuses on critical spatial regions, enables the network to more intelligently screen and integrate valid information from different hierarchical levels. The results of the w/o SFF setting indicate that using DARM to fuse hierarchical features is not well suited to the progressive structure of HFIENet. Moreover, the lower

EN and MI scores further suggest that the fused outputs preserve source image information less effectively compared with the network that incorporates the SFF module. Similarly, the overall performance of the model decreases when the pixel loss function is removed (w/o pixel). Notably, the MI metric increases across all variants. This is because removing core components weakens the network's ability to suppress redundancy and enhance complementarity; instead, the network excessively retains redundant and common information from the source images, leading to an increase in the statistical correlation between the fused image and the source images. However, the accompanying decrease in other metrics reveals hidden degradation in fusion quality, such as a loss of complementary information, reduced contrast, and blurring of key regions. Therefore, based on a comprehensive assessment of metrics including EN and SD, HFIENet achieves optimal performance.

Furthermore, Fig. 9 intuitively demonstrates the feature representations of different modalities before and after complementary attribute enhancement. It can be observed that after the CA stage, the modality-specific features extracted by the residual networks in different branches are significantly enhanced, for instance, the salient targets in infrared images and the background in visible images. The

Table 8

Performance comparison on the MSRS dataset in terms of parameter count, FLOPs, and processing time. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	Parameters (M)	FLOPs (G)	Time (s)
U2Fusion	0.659	366.340	0.123
MFEIF	0.371	924.743	0.108
YDTR	0.107	138.921	0.222
IRFS	0.241	346.544	0.116
DCINN	2.097	17.026	0.031
CrossFuse	21.462	81.193	0.649
DATFuse	0.011	23.698	0.055
TUFusion	19.071	272.992	0.233
LRRNet	0.049	3.022	0.448
Ours	0.519	660.823	0.207

above results fully confirm that the proposed sub-components exhibit strong synergy, which collectively contributes to excellent image fusion quality.

4.5. Computational complexity analysis

To evaluate the efficiency and deployment feasibility of the proposed HFIENet, we compare its computational complexity with that of other representative methods. Three key metrics are considered: model parameters (Parameters, M), floating-point operations (FLOPs, G), and the average inference time per fused image on the MSRS test set (Time, s).

The results are summarized in Table 8. Although HFIENet is not the model with the smallest parameter count, it remains within the category of lightweight models when compared with CrossFuse and TUFusion. Its relatively high FLOPs are mainly attributable to the multi-level feature extraction and fusion architecture. In particular, the cross-attention mechanism and differential information interaction embedded in the Information Exchange (IE) module introduce additional computational demands to fully exploit and integrate complementary features from both modalities. This computational investment, however, is critical to achieving the superior fusion performance of the model. Moreover, HFIENet requires only 0.207 s on average to process a pair of images, delivering competitive inference speed that is fully acceptable for many non-strict real-time application scenarios.

Overall, HFIENet strikes an effective balance between fusion performance and computational cost. While maintaining a relatively compact model size, it achieves practical inference efficiency and demonstrates strong potential for deployment in applications requiring high-quality fused images.

4.6. Downstream IVIF applications

To investigate the contribution of image fusion to downstream tasks, we evaluate the performance of the fusion results on semantic segmentation and object detection.

4.6.1. Semantic segmentation performance

We conduct segmentation experiments using the pre-trained Segformer [43] as the segmentation model. Specifically, we utilize the fused training and test images generated by each method on the MSRS dataset. Each method performs segmentation based on its own fusion results. The segmentation model is trained for 300 epochs, with the backbone network frozen during the first 100 epochs to stabilize learning.

Fig. 10 presents qualitative segmentation results under two lighting conditions. In the daytime scenario, U2Fusion, IRFS, DCINN, DATFuse, and LRRNet fail to segment the “person” within the green bounding box, resulting in a loss of semantic information. The segmentation accuracy for the “person” category remains suboptimal in the results

of MFEIF, YDTR, and CrossFuse. In contrast, only TUFusion and our method are capable of segmenting the “person” category with clearer and more distinctly defined boundaries. In nighttime scenes, reduced illumination impacts visual clarity, introducing instability in segmentation outcomes. For instance, MFEIF, DCINN, CrossFuse, and LRRNet exhibit notable content loss when segmenting the “color cone”. Only IRFS and our method achieve segmentation results that closely match the ground truth for the “color cone”.

The quantitative results of the semantic segmentation experiments are summarized in Table 9. Our method demonstrates superior performance in segmenting categories such as “unlabeled”, “bike”, “car stop”, and “color cone”, and achieves the highest mIoU score among all methods. These results further highlight the effectiveness of our fusion network in enhancing performance on downstream segmentation tasks.

Additionally, we use the fusion results from the M3FD dataset to verify the generalization ability of our model in segmentation experiments. Since this dataset lacks official segmentation labels, we only employ the YOLO11 [44] semantic segmentation model pre-trained on the COCO dataset to conduct a visual comparison of the fusion results generated by different methods. The M3FD dataset contains six categories, among which ‘person’ and ‘car’ are consistent with the categories in the COCO dataset and are the most common. To ensure the reliability of the results, we only focus on the segmentation performance of ‘person’ and ‘car’ in the semantic segmentation experiments, with the results shown in Fig. 11. It can be observed that compared with the segmentation results of the unfused images, the segmentation results of the fused images are generally improved. Relatively speaking, in the results of HFIENet, the segmentation of the vehicle on the left and the person on the right is more accurate, which indicates that our model also exhibits excellent generalization ability in downstream segmentation tasks.

4.6.2. Object detection performance

The M3FD dataset is designed for object detection tasks and contains labels for six object detection-specific categories. We also focus specifically on the detection of “person” and “car”, and use the YOLO11 object detection model pre-trained on the COCO dataset. First, we randomly select 420 image pairs from the M3FD dataset and input them into various fusion methods to generate fused images. These fused images are then directly fed into YOLO11 for detection, with the results shown in Fig. 12 and Table 10.

As can be seen from the figure, for small distant targets, the single-modal images and the images generated by methods such as U2Fusion and MFEIF all have cases where targets are not detected. In contrast, only our method, along with YDTR, DCINN, and LRRNet, successfully detected “person”. This indicates that our fusion results effectively preserve the salient targets and edge features in the source images. In addition, the results in the table show that HFIENet ranks first and second in the detection accuracy of ‘person’ in terms of mAP@0.5 and mAP@0.5:0.95, respectively. Compared with single-modal images, the fused images generated by HFIENet effectively improve the accuracy of visual detection.

5. Conclusion

In this paper, we propose a Hierarchical Feature Information Exchange Network (HFIENet) for infrared and visible image fusion. The framework incorporates an Information Exchange (IE) module that consists of two stages: complementary attribute enhancement and differential information exchange. This module effectively extracts cross-modal complementary features and achieves more comprehensive feature fusion. In the reconstruction phase, the Selective Feature Fusion (SFF) module is introduced to adaptively select fused features across multiple hierarchical levels. Skip connections are also employed to mitigate information loss and enhance the network’s representational capacity. Extensive experiments on publicly available datasets demonstrate that

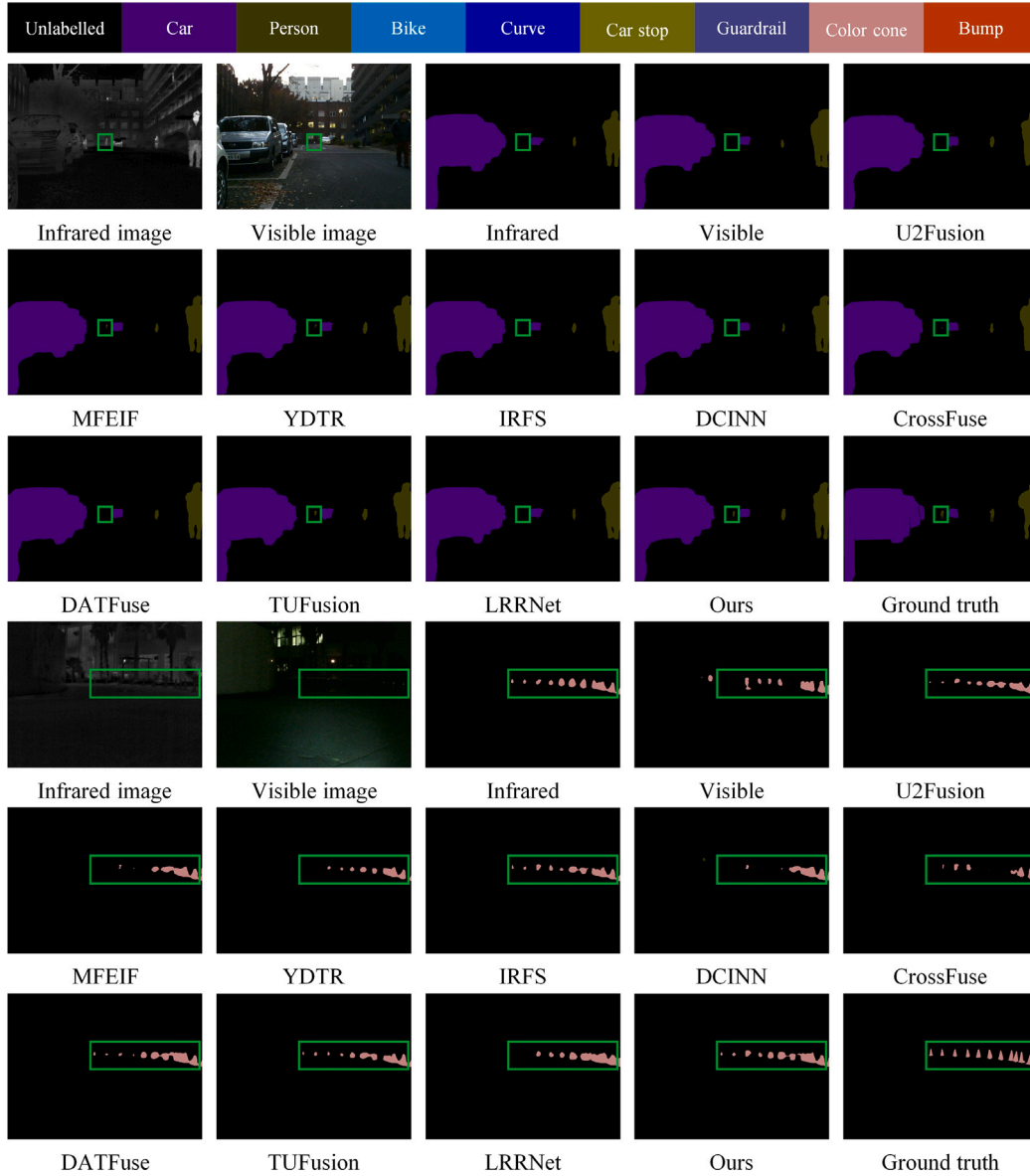


Fig. 10. Qualitative comparison of semantic segmentation between HFIENet and other methods on the MSRS dataset.

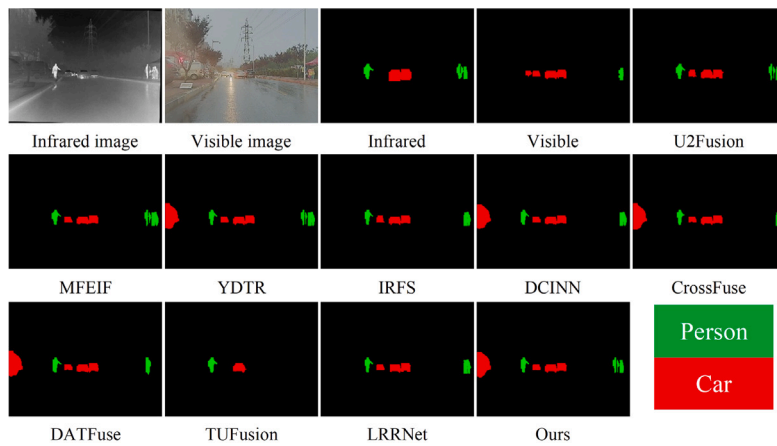


Fig. 11. Qualitative comparison of semantic segmentation between HFIENet and other methods on the M3FD dataset.

Table 9

Quantitative semantic segmentation results of different methods on the MSRS dataset. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	IoU(%)									mIoU(%)
	unlabeled	car	person	bike	curve	car stop	guardrail	color cone	bump	
Infrared	98.04	86.44	70.47	67.03	56.64	64.42	58.78	55.46	69.12	69.60
Visible	98.06	88.24	59.97	69.79	54.43	69.57	73.60	58.23	71.51	71.49
U2Fusion	98.15	87.89	69.38	70.01	54.36	69.47	68.63	55.37	68.32	71.29
MFEIF	98.17	88.03	69.00	68.95	54.77	69.17	74.38	56.11	71.82	72.27
YDTR	98.19	88.12	68.66	69.64	54.45	69.57	73.49	56.32	72.71	72.35
IRFS	98.22	88.45	67.83	69.80	55.27	69.32	70.79	58.86	74.28	72.54
DCINN	98.19	88.07	68.36	68.04	55.46	70.69	75.24	55.58	70.69	72.26
CrossFuse	98.18	88.21	65.73	69.17	56.15	70.41	73.01	57.34	71.17	72.15
DATFuse	98.22	88.02	68.98	69.57	54.02	70.75	69.12	58.62	74.29	72.40
TUFusion	98.15	87.89	67.84	69.66	54.54	70.44	70.35	56.65	72.71	72.03
LRRNet	98.22	88.68	68.22	67.98	57.17	71.47	69.79	58.25	71.44	72.36
Ours	98.24	88.30	69.18	70.04	56.07	71.58	71.92	59.68	72.06	73.01

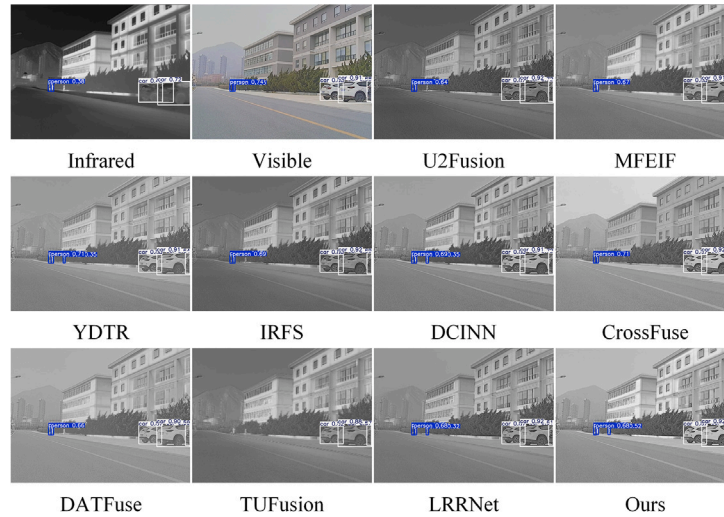


Fig. 12. Qualitative comparison of object detection between HFIENet and other methods on the M3FD dataset.

Table 10

Quantitative object detection results of different methods on the M3FD dataset. The optimal and suboptimal results are highlighted in red and blue, respectively.

Methods	mAP0.5		mAP0.5:0.95	
	Person	Car	Person	Car
IR	0.592	0.669	0.333	0.386
VI	0.489	0.811	0.247	0.546
U2Fusion	0.608	0.812	0.324	0.547
MFEIF	0.594	0.810	0.337	0.541
YDTR	0.605	0.815	0.342	0.548
IRFS	0.596	0.815	0.335	0.556
DCINN	0.610	0.799	0.332	0.529
CrossFuse	0.556	0.821	0.305	0.553
DATFuse	0.571	0.789	0.315	0.507
TUFusion	0.456	0.699	0.264	0.453
LRRNet	0.588	0.818	0.332	0.551
Ours	0.615	0.819	0.340	0.549

HFIENet outperforms existing state-of-the-art fusion methods in both visual quality and quantitative metrics, while exhibiting strong generalization ability. Moreover, the high fusion quality significantly improves the performance of downstream semantic segmentation tasks.

CRediT authorship contribution statement

Xiangcan Du: Writing – original draft, Validation, Software, Investigation. **Kun Qian:** Supervision, Software. **Haojie Zhou:** Validation,

Methodology, Investigation, Funding acquisition. **Hongbin Yu:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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